

Coco (Cyan Robotics) robots specialise in 15-minute food deliveries from local restaurants. Tortoise, on the other hand, debuted mobile vending robots. A vendor can load the Tortoise robot with goods. The robot goes around campuses or street sidewalks ‘selling’ these. Customers can take what they want and self-checkout. The robots are supervised remotely by human operators.

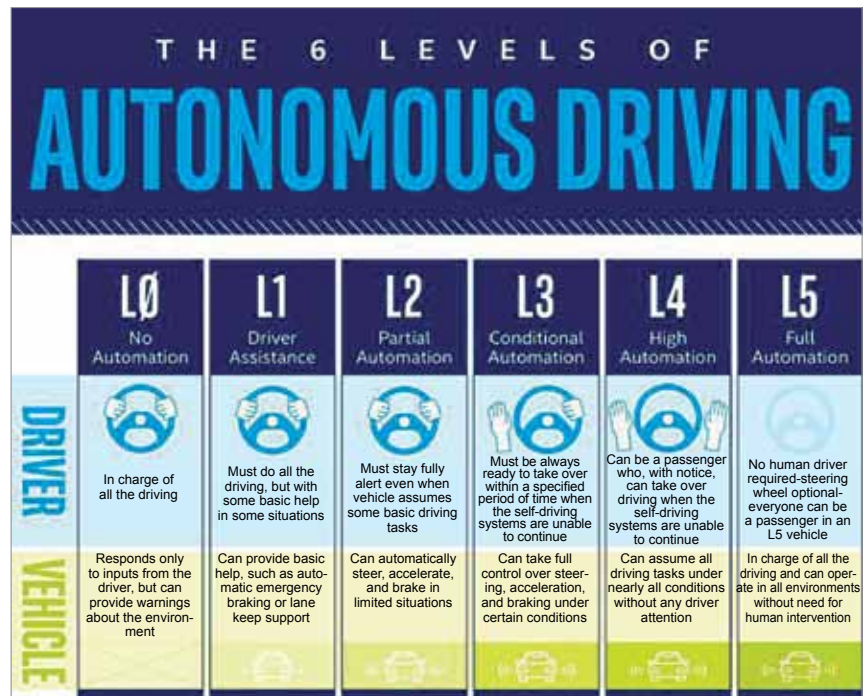
Ottobot from Ottonomy.IO makes life convenient for passengers at airports by delivering food near their respective gates. They operate in several airports, including Cincinnati, Pittsburgh, and Rome. At this year’s CES, Ottonomy.IO revealed Yeti, a self-dispensing robot that can deliver goods even when the customer is not available at home to collect it. It would be risky to deliver fragile objects though, because they slide down from the locker box, and there is a chance of breakage.

Goggo Network is an autonomous mobility solutions provider in Europe. Like most others, they also offer their solutions as-a-service to multiple partners. They have small and big, single- and multiple-stop robots. Last year, they joined the French government funded 5G Open Road project. They operate in France and Spain, with clients like Carrefour and DIA.

Swiss startup LOXO plans to launch its mobile delivery robots in public roads this spring. They plan to start with Switzerland and expand to the UK and Germany. Their AV is essentially a compact, electrically-powered box on wheels with compartments for packages that uses an array of sensors to deliver its self-driving functionality. LOXO travels at typical urban traffic speeds and stops immediately at the first sign of any danger. It is observed remotely by trained personnel who can take control at any time, if necessary, and also speak to the users.

What’s in the bots

Autonomous last-mile delivery bots are very similar to self-driving cars.



The six levels of autonomy in vehicles (Courtesy: Intel)

They require almost all the technology, plus some more. Each bot contains a variety of sensing technologies, such as state-of-the-art thermal cameras, point cloud sensors, lidar and radar, inertial ultrasonic sensors, and more. Then, there are processing units, which include central processing units, graphics processing units, neural network accelerators, field-programmable gate arrays, and embedded integrated circuits.

That said, software forms the backbone of any autonomous system. “We have a massive software system of millions of lines of code, both in the robot and in the cloud. In simple terms, each robot has an awareness system. It can see what’s going on, avoid objects, detect objects, brake, accelerate, turn corners. It identifies where it is and signals to central locations if there’s a problem with the device. And the cloud software handles things like payments, route planning, fleet orchestration. It monitors the robot health and energy charge and ensures that the correct number of robots arrive at the

right time at the right locations. It’s a choreography of things in the cloud as well as things on the ground,” explains Alastair Westgarth, CEO of Starship Technologies, in a Forbes interview.

The robots almost always have a data connection, either cellular or Wi-Fi, depending on the area of deployment. They use this to communicate with the cloud. But, in rare cases, if the connection fails, the robots are not stumped. They keep moving safely in permitted routes, till they get reconnected.

During ‘rest time’ the robots usually move to a hub for charging and software upgrades. This is when diagnostics are uploaded to the cloud. Companies use this to study how the robot’s day went, and to improve its behaviour further.

Many technologies, from perception to path planning and mapping, appear common to autonomous delivery robots and autonomous cars. But, Ritukar Vijay, founder and CEO of Ottonomy.IO explains in one of his articles that, “Although the delivery robots are low in speed, they often engage in more complicated, unstructured scenarios and